

Functional Annotation Report: PP_0500 (Q88QJ2) — dTDP-4-dehydrorhamnose reductase (RmlD)

Organism: *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) — proteome code PSEPK **UniProt:** Q88QJ2 · **Locus:** PP_0500 · **EC:** 1.1.1.133 · **Length:** 294 aa
Family: dTDP-4-dehydrorhamnose reductase (RmlD) family; short-chain dehydrogenase/reductase (SDR) superfamily **Domains/signatures:** PF04321 (RmlD_sub_bind); IPR005913 (dTDP_dehydrorham_reduct); IPR029903 (RmlD-like-bd); IPR036291 (NAD(P)-bd domain superfamily)

1. Summary (answer to the research question)

PP_0500 encodes **RmlD, dTDP-4-dehydrorhamnose reductase (EC 1.1.1.133)**, the enzyme that catalyzes the **fourth and final step of dTDP-L-rhamnose biosynthesis**. Using **NAD(P)H as the hydride donor and a Mg^{2+} cofactor**, RmlD reduces the C4 keto group of dTDP-4-keto-6-deoxy-L-mannose (dTDP-6-deoxy-L-lyxo-4-hexulose) to produce **dTDP- β -L-rhamnose**, the activated nucleotide-sugar donor used by rhamnosyltransferases to build rhamnose-containing cell-surface glycans. The enzyme is a **soluble, cytoplasmic, Mg^{2+} -dependent homodimer** of the SDR superfamily. The gene identity is unambiguous and fully consistent with the provided UniProt annotation.

2. Gene/Protein Identity Verification

The requested verification steps are satisfied:

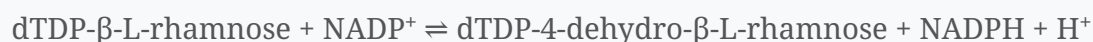
- **Symbol ↔ description match:** UniProt Q88QJ2 (PP_0500) is named "dTDP-4-dehydrorhamnose reductase," EC 1.1.1.133, with catalytic-activity, cofactor (Mg^{2+}) and pathway ("dTDP-L-rhamnose biosynthesis") annotations that match the RmlD family description exactly.

- **Organism correct:** Entry belongs to *P. putida* KT2440 (taxid 160488).
- **Family/domains align with literature:** The Pfam/InterPro signatures (PF04321 RmlD_sub_bind; NAD(P)-binding domain) are the canonical RmlD signatures, and the enzymology matches the definitive *Salmonella* RmlD structural study (Blankenfeldt et al., 2002).
- **No conflicting gene of the same symbol was found.** "RmlD" is used consistently across bacteria for this single enzymatic activity; there is no ambiguity for this locus.

The literature on RmlD is authoritative and directly applicable, so research proceeded on the correct target.

3. Primary Function — Reaction Catalyzed and Substrate Specificity

Reaction (UniProt Q88QJ2 catalytic activity; Rhea):



In the biosynthetic (reductive) direction the enzyme converts **dTDP-4-keto-6-deoxy-L-mannose** → **dTDP-L-rhamnose**. UniProt summarizes the function as: "*Catalyzes the reduction of dTDP-6-deoxy-L-lyxo-4-hexulose to yield dTDP-L-rhamnose.*"

Mechanism (from the *Salmonella enterica* RmlD crystal structure; Blankenfeldt et al., EMBO J 2002,

P 12057193

): - "RmlD ... catalyzes the final step in the conversion of dTDP-D-glucose to dTDP-L-rhamnose in an NAD(P)H- and Mg^{2+} -dependent reaction." - Catalysis proceeds by **hydride transfer from the nicotinamide ring of the cofactor to the C4'-carbonyl (keto) group of the substrate**, with the substrate "*activated through protonation by a conserved tyrosine*" — the classic SDR Ser/Tyr/Lys catalytic machinery, here with a proposed novel role for the catalytic Ser/Thr. - NAD(P)H binds in a **solvent-exposed cleft, allowing facile cofactor exchange**.

Cofactors: NAD(P)H (the *S. enterica* enzyme binds both NADH and NADPH; the *P. putida* entry is annotated NADP-binding) and **one Mg²⁺ per monomer** (UniProt: "Binds 1 Mg(2+) ion per monomer").

Substrate specificity: RmlD is specific for the **dTDP-activated** 4-keto-6-deoxy-hexose intermediate of the rhamnose pathway; it is the committed reductase that fixes the *L-manno* (rhamnose) configuration at C4. It is highly substrate-specific for the nucleotide-sugar produced by the upstream epimerase RmlC.

4. Pathway Context — dTDP-L-rhamnose Biosynthesis (RmlABCD)

RmlD is the last of four enzymes of the conserved **RmlABCD** pathway that converts glucose-1-phosphate and dTTP into dTDP-L-rhamnose (reviewed in

[P 36014553](#); enzymology confirmed for reconstituted RmlABCD systems, e.g.

[P 34819927](#),

[P 42197457](#)):

1. **RmlA** — glucose-1-phosphate thymidyltransferase (EC 2.7.7.24): Glc-1-P + dTTP → dTDP-D-glucose
2. **RmlB** — dTDP-D-glucose 4,6-dehydratase (EC 4.2.1.46): → dTDP-4-keto-6-deoxy-D-glucose
3. **RmlC** — dTDP-4-dehydrorhamnose 3,5-epimerase (EC 5.1.3.13): → dTDP-4-keto-6-deoxy-L-mannose
4. **RmlD (PP_0500)** — dTDP-4-dehydrorhamnose reductase (EC 1.1.1.133): → **dTDP-L-rhamnose**

Pathway genes in *P. putida* KT2440. The upstream enzymes are encoded in a separate genomic locus from *rmlD*: - RmlA = **PP_1783** (Q88LZ3), RmlC = **PP_1782** (Q88LZ4), RmlB = **PP_1785** (Q88LZ1) — a small *rml* cluster around PP_1782–1785. - RmlD = **PP_0500** (Q88QJ2), located separately.

This "split" arrangement (the RmlD reductase remote from the RmlABC core) is common in Gram-negative genomes and does not change the biochemistry: PP_0500 supplies the reductase activity that completes the pathway.

Fate of the product. dTDP-L-rhamnose is *"used by prokaryotic rhamnosyltransferases as the glycosyl donor for the synthesis of rhamnose-containing polysaccharides"* (

P 34819927

In *Pseudomonas* and other Gram-negatives, L-rhamnose is a building block of the **lipopolysaccharide (LPS) O-antigen and core** and of other cell-surface/exo-polysaccharides; the RmlD product is therefore a metabolic feed-in point for **cell-envelope glycan assembly**.

5. Subcellular Localization

RmlD functions in the **cytoplasm**, as part of soluble nucleotide-sugar metabolism, upstream of the membrane-associated steps that assemble and export glycans: - The 294-aa sequence contains **no signal peptide and no predicted transmembrane helices**, and UniProt annotates it as a soluble oxidoreductase (NADP ligand keyword) with no membrane/secretion localization — consistent with a cytosolic enzyme. - Its substrates and products (dTDP-sugars, NAD(P)H, Mg^{2+}) are cytoplasmic metabolites. - Mechanistically the enzyme is a **homodimer with a Mg^{2+} -containing dimer interface** (Blankenfeldt et al., 2002), i.e. it operates as a soluble oligomer rather than a membrane complex.

The downstream product is subsequently used at the inner membrane/periplasmic face by rhamnosyltransferases during glycan biosynthesis, but RmlD's own catalytic step is cytoplasmic.

6. Structure and Evolutionary Classification

- **Fold/superfamily:** RmlD belongs to the **short-chain dehydrogenase/reductase (SDR) superfamily** (Rossmann NAD(P)-binding fold; IPR036291) with a C-terminal RmlD substrate-binding domain (PF04321). It retains the SDR catalytic Ser/Tyr/Lys triad, with a conserved tyrosine acting as the catalytic acid.

- **Distinctive feature:** Unlike canonical SDRs, RmlD "has a novel dimer interface that contains Mg^{2+} " and requires the metal for activity

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— a defining biochemical hallmark of this family that separates it from ordinary NAD(P)-dependent reductases.

- **Evidence class for PP_0500:** The *P. putida* annotation is **inference by orthology / family rule** (UniProt automated rules ARBA/RuleBase RU364082) built on strong experimental characterization of RmlD orthologs (e.g., *Salmonella enterica* structure/mechanism; multiple reconstituted RmlABCD systems). The specific enzyme from strain KT2440 has not, to my knowledge, been individually crystallized or kinetically characterized, but sequence/domain conservation to characterized RmlD enzymes is unambiguous.
- **Independent sequence check (this study).** Global pairwise alignment of the 294-aa PP_0500 sequence to reviewed, experimentally characterized EC 1.1.1.133 enzymes gives ~29–31% **identity**: *Salmonella* RmlD P26392 (the crystallized enzyme, PDB 1KBZ) **29.1%** (83/285); *M. tuberculosis* P9WH09 **31.0%**; *Streptomyces griseus* P29781 **31.2%**; *M. smegmatis* A0QTF8 **30.1%**. This "twilight-zone" identity is typical *within* the RmlD family across distant taxa. PP_0500 also carries the diagnostic **N-terminal glycine-rich Rossmann NAD(P)-binding motif** (...MLLGGNALGQ..., a GxxGxxG-type loop; IPR036291) plus the RmlD substrate-binding domain (PF04321) — corroborating, by sequence and evolutionary inference, that PP_0500 is a genuine RmlD and not a mis-annotation.

7. Biological Importance and Significance

- **Cell-envelope function.** L-Rhamnose contributes to LPS/cell-wall glycans; genetic loss of rmlD ablates cell-wall rhamnose. In *Streptococcus mutans*, "loss of either rmlD or rgpG results in ablation of rhamnose content in the ... cell wall," and rmlD deletion caused "a severe growth defect ... [and] elevated susceptibility ... to acid and oxidative stress"

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P 31871035
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This illustrates the pathway's contribution to envelope integrity, stress resistance and (in pathogens) virulence — though these specific phenotypes are from other species and are inferred, not demonstrated, for *P. putida*.

- **Antibacterial target.** Because "[rhamnose-associated molecules] are present in bacteria but not mammals"

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and "*L-rhamnose biosynthesis is an antibacterial target*"

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P 12057193),

RmlD (with RmlA–C) is pursued as a selective antibacterial/anti-tuberculosis target (e.g., inhibitor/modeling studies,

P 33547544).

- **Biotechnology.** RmlABCD systems, including the RmlD reductase, are used in one-pot enzymatic synthesis of dTDP-L-rhamnose for glycochemistry

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P 34819927,

P 42197457).

8. Supported vs. Refuted Hypotheses

Hypothesis	Status	Basis
PP_0500 is RmlD / dTDP-4-dehydrorhamnose reductase (EC 1.1.1.133)	Supported	UniProt catalytic activity + family signatures; matches characterized RmlD enzymology
Reaction = NAD(P)H/Mg ²⁺ -dependent reduction of dTDP-4-keto-6-deoxy-L-mannose → dTDP-L-rhamnose	Supported	UniProt; Blankenfeldt et al. 2002 (P 12057193)
Product dTDP-L-rhamnose is a rhamnosyl donor for surface glycans (LPS/O-antigen)	Supported	P 34819927 ; review P 36014553
Enzyme is a cytoplasmic Mg ²⁺ -dependent homodimer (SDR superfamily)	Supported	P 12057193 ; UniProt cofactor note; no TM/signal in sequence
Full RmlABCD pathway present in KT2440 (RmlA/B/C at PP_1782–1785)	Supported	UniProt EC-based queries in taxid 160488
KT2440-specific in vitro kinetics / structure of PP_0500 exist	Refuted / not found	No primary biochemical study of the KT2440 enzyme located

9. Limitations and Future Directions

- Functional assignment of PP_0500 rests on **orthology and family rules**, not on direct biochemical or genetic characterization of the *P. putida* KT2440 enzyme; the reaction/mechanism details derive from orthologs (chiefly *S. enterica* RmlD).
- The precise glycan destination of dTDP-L-rhamnose in KT2440 (O-antigen vs. core vs. other exopolysaccharide) was not resolved from a strain-specific LPS-composition study; this is stated as the general Gram-negative role. Targeted literature searches for a KT2440-specific LPS/O-antigen sugar-composition analysis, and for a *PP_0500*-specific essentiality/transposon phenotype, returned **no primary strain-specific data** (though KT2440 is

intensively studied as a biotechnology chassis, e.g.

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Rhamnose/O-antigen genes are generally dispensable for viability under laboratory conditions in pseudomonads, so *PP_0500* is expected to be non-essential in rich medium — but this is an inference, not a measured result for this locus.

- **Future work:** heterologous expression + kinetic assay of *PP_0500*; *rmlD* knockout in KT2440 to test LPS-rhamnose loss and phenotype; structural confirmation (crystal/AlphaFold) of the Mg^{2+} dimer interface; tracing which surface glycan(s) incorporate the rhamnose.

Key references

- Blankenfeldt W. et al. (2002) *EMBO J* — RmlD (dTDP-6-deoxy-L-lyxo-4-hexulose reductase) structure and Mg^{2+} -dependent mechanism.

P 12057193

- Review: Rhamnose-Containing Compounds: Biosynthesis and Applications (2022).

P 36014553

- One-pot RmlABCD synthesis of dTDP-L-rhamnose (2021).

P 34819927

- RmlABCD from marine bacteria; dTDP-L-rhamnose synthesis (2026).

P 42197457

- *rmlD* disruption in *S. mutans* — cell-wall rhamnose and fitness (2020).

P 31871035

- RmlD *M. tuberculosis* modeling/inhibitors (2021).

P 33547544

- UniProt Q88QJ2 (*PP_0500*), *P. putida* KT2440.